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► To cite this version:

José Ignacio Carrasco, Ognjen Obućina. The dynamics of remittance behaviour among Senegalese men and women in Spain. *International Migration*, 2023, 61 (1), pp.239-255. <10.1111/imig.12997>. <hal-03608487>

HAL Id: hal-03608487

<https://hal.science/hal-03608487v1>

Submitted on 14 Mar 2022

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The dynamics of remittance behaviour among Senegalese men and women in Spain

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Funding information

Chilean National Agency of Research and Development (ANID), grant "Programa Formación Capital Humano Avanzado"

Abstract

This article explores the conditions under which Senegalese immigrants in Spain send remittances home, beginning with the premise that remittances are intertwined with migration histories and migrants' incorporation into host societies. Given the strong gender norms in Senegal, we perform separate analyses for men and women. We use a longitudinal approach to analyse how remittance behaviour is affected by immigrants' characteristics, their economic integration, and their ties to origin and destination. Our data come from the *MAFE* and *MESE* surveys, which were implemented in 2008 and 2011, respectively. The results indicate that remitters constitute a clear majority among Senegalese immigrants in Spain. The Kaplan–Meier analysis shows that they rarely stop remitting once they start doing so, and the multivariate analysis reveals a strong positive association between employment and remittance sending. Although most coefficients in the full model are similar for men and women, some important differences emerge as well.

KEYWORDS

labour market incorporation, longitudinal, remittances, Senegal, Spain

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INTRODUCTION

Migrant remittances have grown steadily over recent decades. Not only have they surpassed official development assistance in low-to-middle income countries, but they were also larger than foreign direct investments in 2019 (World Bank, 2020). Although remittances to Africa constitute only 10 per cent of total remittances compared to other regions such as Asia and Latin America, they constitute a significant share of the GDP for some African countries (OECD, 2005). In Senegal, official monetary remittances accounted for 10 per cent of their 2011 GDP (Cisse, 2011), and a similar figure (10.7 per cent of GDP) is observed for 2020 (UNDP, 2021). Research on the impact of remittances on development indicates that remittances have contributed to a 30 per cent reduction in Senegalese poverty (Diagne & Diane, 2008). For scholars and policymakers, however, the importance of remittances extends beyond their economic dimension, as remittances and their determinants are intertwined with migration histories and migrants' incorporation into host societies, and they are arguably one of the central manifestations of immigrants' transnational ties. Remittances help sustain pivotal contemporary migration practices such as transnational families and circular migration, and they can also encourage further migration flows (de Haas, 2010; IFAD, 2017). By drawing on the previous empirical evidence of Senegalese migrants in Europe, this article focuses on the Spanish context and aims to disentangle the way in which migrant remittance behaviour is affected by individual characteristics (e.g., gender and education), economic integration (e.g., employment status), and their ties to origin (e.g., family reunification). Our main research question is: Under what conditions do Senegalese in Spain send remittances home? We address this question by using a unique dataset that combines two retrospective surveys, namely *Migration between Africa and Europe* (MAFE, conducted in 2008) and *Migrations Between Senegal and Spain* (MESE, conducted in 2011).

Although remittance behaviour among Senegalese in Europe has already been addressed in some recent studies (among others, Chort et al., 2012; Vickstrom & Beauchemin, 2016), we complement the previous research in four ways. First, we introduce a dynamic component into the analysis of remittances by looking at when Senegalese immigrants start remitting and for how long they continue to do so. Second, we focus on a single destination country: Spain. This can be of importance when working with small samples, as our results cannot be biased by unobserved heterogeneity due to selection into different destinations. Third, we introduce into our model a wide range of variables reflecting family and economic ties to the home and destination countries. Fourth, we add a gender perspective to the overall analysis of remittance sending among Senegalese migrants by performing separate analyses for men and women. Given the main features of the paper, it can be argued that its contribution to the literature on remittances is not confined strictly to the context of Senegalese migration. In particular, this concerns the longitudinal nature of the study, as a large majority of studies on remittances have been cross-sectional. Additionally, although the gender dimension of remittances is not entirely absent from the literature, it has been rarely addressed in previous studies.

SOCIAL CONTEXT OF THE STUDY

Senegalese migration to Europe

With a population of around 17 million people, Senegal is a country characterized by low economic development and relative political stability. Although an important destination for some neighbouring West African countries, Senegal is also a country with a high level of emigration. According to UN-DESA (2019), around 45 per cent of Senegalese migrants live in another African country (with the most important destinations being Gambia and Mauritania), whereas 48 per cent of Senegalese migrants have opted for a European country. In the latter group, France, Italy, and Spain are by far the most important destinations, where the Senegalese populations are, respectively, 130,000, 104,000, and 52,000. Looking at the trends since 2010 (which roughly coincides with the time

of the survey upon which this study is based), the UN DESA data suggest that the presence of Senegalese has moderately increased in France and Italy while it has very slightly decreased in Spain. In spite of this trend, the Senegalese in Spain still rank as the second largest group of African immigrants in the country (after Moroccans), followed by Algerians and Nigerians (INE, 2021).

While France has long been a typical destination country for Senegalese migrants due to its former colonial links and language ties, Spain and Italy have recently become attractive destinations. The tourism and industry sectors attracted migrants to Italy during the 1990s, whereas Spain became important as a destination country due to a demand for labour in construction and the agricultural sector (Baizán & González-Ferrer, 2016). In his study on the occupational attainment of Senegalese migrants in France, Italy, and Spain, Obućina (2013) showed that labour pathways follow a U-shaped trajectory. This means that, on average, the occupational status of Senegalese migrants drops upon arrival in Europe and then increases over duration of stay, although with significant differences by destination. The pan-European approach in this study is complemented by Castagnone et al. (2015), who find that, in contrast to those residing in France and Italy, hardly any Senegalese in Spain are employed in skilled occupations even ten years following their immigration. It is likely that compositional factors underly these differences, as the same study finds that Senegalese migrants in Spain differ from those in France and Italy, namely in that they arrived in Europe more recently, are younger, and have lower levels of education. In this sense, questions naturally arise in regard to what extent Senegalese migrants are able to accumulate enough economic capital to start and maintain a remittance flow (Bruzzone et al., 2006).

Gendered patterns of Senegalese migration

Gender is one of the most basic elements structuring Senegalese migrants' experiences in Europe, as Senegalese men and women differ considerably in terms of the volume and type of migration. Although previous research suggests that migration aspirations in Senegal are nearly as widespread among women as among men (Carling et al., 2013), Senegalese migration is still highly influenced by a traditional patriarchal context that discourages female migration. Moreover, it is possible that remittances themselves are one of the mechanisms that shape the gendered nature of Senegalese migration. For example, Beauchemin et al. (2015) argue that women are expected to stay in Senegal in order to ensure that the migrant husband will fulfil his obligations towards the home community. The process of feminization of migration in Senegal is therefore slow, and international migration is still largely considered a masculine activity (Flahaux et al., 2014; Vause & Toma, 2015). It should, however, be noted that female family members remaining in Senegal often participate in the Senegalese culture of migration as "active stayers", both by mobilizing financial resources to support the potential migrant before migration and by engaging in transnational activities after migration (Mondain & Diagne, 2013). When they do migrate abroad, Senegalese women most often do so for family reunification, that is, in order to join their partners already established abroad. However, the share of autonomous migrants among female migrants is not negligible either (Toma & Vause, 2013, 2014). Traditional views about gender roles also become manifest in the economic activity in the destination country. Castagnone et al. (2015) find that Senegalese labour market participation was higher among men than among women across all prominent European destinations, namely France, Italy, and Spain.

THEORETICAL BACKGROUND

In this paper, remittances are understood as transnational economic activities that keep migrants connected to their countries of origin and emerge from a combination of the migrant's capacity and desire to engage in a wider range of transnational practices, such as supporting future migrants, investments, and circular migration (De Haas, 2010; Portes et al., 2002). Our frame of reference draws from Carling (2008) in that we seek to understand the

capacity and desire to remit by studying the characteristics of a particular migration context under study, the nature of families and households, and the demographic dynamics underlying the variance in remittance flows. While economic integration refers to local capacities, the desire to remit is influenced by ties to the country of origin. Also in line with the author, we account for the preconditions for sending remittances (e.g., presence of potential remittance receivers in the origin country), assuming that certain demographic and kinship variables reflect stages in family migration histories that constitute key micro-level determinants of remittance flows.

Family links hypothesis

Family ties are one of the key dimensions determining the level of migrants' attachment to their country of birth. The presence of family members in the origin country can trigger remittance sending through altruistic motivations. However, as migrants derive utility from their family's utility, the association between sending remittances and having family members such as parents or siblings in Senegal may not be exclusively motivated by pure altruism (Lucas & Stark, 1985). Previous literature has established that remittance sending may also be motivated by contractual arrangements between the migrant and the family members back home (Rapoport & Docquier, 2006). These mostly informal arrangements may concern a wide range of transactions between the migrant and the family. For instance, some migrants send money back home to compensate for the care of children left behind. Other motivations for remittance sending include a desire to maintain access to family property or financial support for the future migration of other family members. Bearing in mind the importance of both altruistic motivations and contractual arrangements – and assuming that they are not mutually exclusive – we expect that, all else being equal, family ties to Senegal will be positively associated with remittances (*the family links hypothesis*).

Just as our data allow us to identify family members living in Senegal each year after migration, we also have information on partners, children, and siblings living in Spain. Since family ties at the destination end of the migration process can affect remittance behaviour as well (Menjivar et al., 1998), we incorporate these variables into our multivariate model. A stronger presence of family members in the host country may cause migrants to shift their priorities, thereby weakening ties to the country of origin and having a negative effect on remittances. Furthermore, as some family members in the destination may need to be supported financially, this limits the resources available for sending remittances. Viewed from this perspective, a stronger presence of family members lowers the likelihood of remittance sending. On the other hand, if living together with a partner in Spain is the alternative to having a transnational family with a Senegal-based partner, this could result in a higher standard of living and more capacities to remit. Further, similar to the role of a strong presence of peer migrants and the migrant community (Chort et al., 2012), a stronger presence of family members and relatives may also imply stronger social control over immigrants' ties to their country of origin. This may result in a higher likelihood of remittance sending. As this paper does not specify hypotheses on the link between family ties in the destination country and remittances, this part of the analysis is strictly exploratory.

Material and physical links

Senegalese emigrants and returnees are more likely than non-migrants to invest in land, housing, and businesses in the home country (Mezger & Beauchemin, 2015). As owning property in Senegal creates practical and symbolic attachment (Obeng-Odoom, 2010), the investments can be an important way to maintain ties with the origin country. However, maintaining these investments requires resources that are often provided by remittances, and most previous studies (e.g., Lucas & Stark, 1985) indeed find a positive association between property in the origin country and remittance sending. In line with these considerations, we also expect to find that owning assets in Senegal increases the likelihood of remittance sending (*the material links hypothesis*). Visits to the home country are

another important instrument for sustaining ties to the country of origin. Even short visits can help sustain emotional links to the origin (Cai, 2003), and most previous studies find that traveling to the home country or family members visiting the destination country is positively associated with the decision to remit (Cai, 2003; Vickstrom & Beauchemin, 2016). Because of this, we expect our study to also find a positive association between visits and remittance sending (*the physical links hypothesis*).

Migration decision-making and remittances

Drawing on the new economics of labour migration (NELM) theory, Lucas and Stark (1985) suggested that remittances form part of a family adaptation strategy to reduce economic risk through some of its members migrating, thus premising that remittance decisions are linked with migration decisions. Migration itself should thus be understood as risk aversion behaviour, in which the household rationally decides to send some of its members to places where the labour market has different characteristics ("non-correlated labour markets"). Under this strategy, migrants receive assistance for their journey to new destinations and during unemployment periods while overseas. In exchange, households experiencing economic hardship would expect to receive remittances. Regarding household risk diversification, evidence shows a positive significant association between the presence of a family adaptation strategy and remittance sending (Cai, 2003). Therefore, from the perspective of the NELM theory, we expect to find a negative association between agency in migration decision-making and remittance sending. Put another way, it is expected that individuals who autonomously chose to migrate will have a lower likelihood to remit (*the autonomous migrant hypothesis*). We should, however, point out that this hypothesis mainly concerns male migrants, for reasons discussed below.

Economic integration and capacities to remit

Turning to the factors at work at the destination, socio-economic integration influences migrants' capacity to remit, as resources can be sent to the origin country through the provision of reduced constraints and access to the labour market. In the words of Hammond (2014): "*The more settled people are in terms of their immigration status, their housing, educational achievement, employment, the more likely they are to be able to engage meaningfully in transnational activities*" (p. 1013). Employment status is generally positively associated with the likelihood to remit (Carling & Hoelscher, 2013; Castagnone et al., 2015). In their Norwegian study, Carling and Hoelscher (2013) find that immigrants with secure employment are more likely to remit than are those with insecure or no employment. Using a cross-sectional approach, Diagne and Rakotonarivo (2010) show that employment status is positively associated with remittance sending by Senegalese migrants in France, Italy, and Spain. Amuedo-Dorantes and Pozo (2006) show that the type of employment and occupational sector also matters, with their results indicating that self-employed migrants are less likely to remit than salaried migrants, whereas immigrants employed in the agricultural sector remit more than their counterparts in other labour market sectors. In sum, higher levels of economic integration should be associated with greater economic resources and higher capacity to send remittances (Carling & Hoelscher, 2013). Therefore, we expect that employed migrants are more likely to remit (*the capacity hypothesis*).

Gender-remittance nexus

Although interest in the so-called "gender-remittance nexus" is growing, the findings of studies are rarely consistent because the results vary substantially across social contexts (King et al., 2013). However, some

patterns do emerge when looking at gender differences in motives to remit. A number of studies find that women's remittance behaviour is more often motivated by family ties, whereas men are more often driven by economic motivations. Despite the possible differences between men and women in the magnitude of coefficients, we expect most principal associations of interest (which are formulated in our hypotheses) to be the same for men and women. A notable exception, however, concerns the links between remittances and agency in choosing to migrate. The migration decision hypothesis implicitly assumes that if migrants do not autonomously decide to leave Senegal, then they migrate out of compliance with a decision made by their (extended) family or local community. This assumption, which is heavily influenced by the NELM theory, may indeed hold for men. Like most countries in West Africa, Senegal is a heavily male-dominated society, which has two important consequences that need to be considered. First, as in most patriarchal settings, the dominant view is that women's migration is riskier and that women are more exposed to dangers in the migration process. Second, when women do migrate, they most often do so as "tied movers", meaning that they either migrate together with their husbands or join them once their husbands are established in the destination country (Toma & Vause, 2013). Therefore, the link between the nature of migration decision-making and remittances is less straightforward among women. Although we can safely assume that women who autonomously decide to migrate will have more agency in deciding whether or not to remit, it is difficult to predict — all else being equal — whether they will actually send money more often or less often when compared to migrating women who are "tied movers".

Duration of stay and remittance flows

The association between remittances and duration of stay in the destination country is complex, due to the interplay of origin country conditions with the capacity and willingness to send remittances. The so-called remittance decay hypothesis predicts that remittances will decrease with duration of stay in the destination country. As more time passes following migration, ties to the origin country may weaken as emotional attachment wanes and the number of potential recipients at origin decreases due to chain migration and older family members dying. Furthermore, remittances intended for paying back loans to finance migration may be restricted to shorter periods of time. Jones (2020) finds that Bolivian emigrants' remittances already begin to decline after two years abroad, which is faster than the decline in other manifestations of transnationalism (such as phone calls or return intentions). However, waning ties with the country of origin may be counterbalanced by an increasing capacity to remit (Carling & Hoelscher, 2013). The interplay of mechanisms working in opposite directions sometimes produces opposing patterns among different groups living in the same geographical context. Arun and Ulku (2011) find that the remittances of Indian and Pakistani immigrants in the Manchester area decrease with time spent in the destination country, while the opposite is the case with Bangladeshi immigrants. On a similar note, Brown (1998) shows that the remittances of Samoan immigrants in Australia decrease considerably with the length of stay, whereas they increase among Tongans during the first fifteen years. However, bearing in mind the simultaneous impact of the forces that encourage remittances, those that impede them, and the varying importance of these forces across time, we find these factors to be reflected in the most common results of previous studies: a non-linear inverse U-shaped relationship between remittances and duration of stay. In other words, remittances increase for some time after arrival, then start decreasing (among others, Cai, 2003; Carling & Hoelscher, 2013; Lucas & Stark, 1985).

DATA AND METHODS

MAFE and MESE surveys

The data used in this paper come from two closely related research projects: *Migrations between Africa and Europe* (MAFE); and *Migrations Between Senegal and Spain* (MESE). The MAFE biographical survey was carried out in five European countries (France, Italy, Spain, the UK, and the Netherlands) and three African countries (Senegal, DR Congo, and Ghana). The survey includes around 600 Senegalese immigrants living in Spain, Italy, and France. In this paper, we include only around 200 respondents who lived in Spain at the time of the survey, which was in 2008. The MESE survey, carried out in 2011, enriches our sample with another 405 Senegalese immigrants in Spain and, unlike the MAFE survey, includes only one origin country (Senegal) and one destination country (Spain). The two surveys are implemented using the same sampling techniques and almost identical questionnaires, so that they can be easily combined into a single dataset (see the study by Kraus, 2019). The surveys include Senegalese-born immigrants aged 25 to 75 years at the time of the survey and who emigrated from Africa at the age of 18 or older (Beauchemin, 2015). The Spanish municipal population register (*Padrón*) served as a sampling frame, which allowed researchers to include a random sample of both documented and undocumented migrants. In the questionnaire, the respondents are asked if there have been periods in their life when they regularly sent money to someone in a country other than the country where the respondent lived. If the answer is positive, the respondents are additionally asked to specify the years during which they sent remittances as well as the country to which they sent money.

Methodology

Before implementing the principal descriptive and multivariate analyses, we transformed the data into person-years. Remittance behaviour is measured through a dichotomous variable that indicates whether the immigrant sends money back to Senegal in the year of observation. The variable takes value 1 in the years in which he or she sends remittances, otherwise it takes value 0. Note that a variable registering the amount of remittances sent was not available in the datasets. Apart from basic trends on remittance behaviour, our descriptive analysis is also based on event history analysis. In the first step, we perform Kaplan–Meier analysis to explore how long it takes immigrants to start remitting. In the second step, we use the same method to analyse how likely immigrants are to stop remitting. The descriptive analysis takes into account post-stratification weights in order to reduce the sampling error. Given that our sample is relatively small and that most immigrants in our sample were recent immigrants at the time of the two surveys, the descriptive analysis of remittance behaviour largely focuses on the first ten years in Spain. Since substantial gender differences can exist in migration behaviour, immigrant integration, and maintenance of ties to origin, all the analyses are performed separately for men and women.

The multivariate analysis is based on a two-level mixed-effects linear probability regression. The advantage of linear probability models, as compared to logistic regression, is an easier interpretability of coefficients, but also the comparability of its estimates across groups (see Mood, 2010 for more details). This may be of a particular importance in this study, given that we perform separate estimates for men and women. The use of the linear probability model implies that the coefficients obtained in this analysis show how much the probability to be remitting in the year of observation changes, in terms of percentage points, when the corresponding independent variable changes by one unit. Our data have a two-level hierarchical structure: the measurement occasions i form level 1, and they are nested within individuals j , who form level 2. Using of mixed effects allows us to estimate both between-individual variance and within-individual variance. Additionally, to obtain a more fine-grained picture of the association between remittance behaviour and time spent in the destination country, the regression also

estimates between-individual slope variances for duration of stay in Spain and its squared term. Our model can thus be written as follows:

$$Y_{ij} = \beta_{0j} + \beta_{1j}t_{ij} + \beta_{2j}t_{ij}^2 + \gamma_n X_{nij} + e_{ij},$$

where:

$$\beta_{0j} = \beta_0 + u_{0j}$$

$$\beta_{1j} = \beta_1 + u_{1j}$$

$$\beta_{2j} = \beta_2 + u_{2j}$$

The overall intercept, averaged across all individuals, is denoted by β_0 , whereas u_{0j} stands for the individual-specific random intercept. β_1 and β_2 are the average slopes of the regression of y on duration of stay in Spain t and its squared term t^2 , respectively. The random slopes, specific to individual j , are denoted by u_{1j} and u_{2j} . X is a vector of n independent variables, whereas e_{ij} is an occasion-specific residual that captures of unmeasured time-varying characteristics. The unstructured covariance matrix allows all variances and covariances of the random effects to be distinct. We obtain robust standard errors by using the cluster option in the regressions.

In line with the model presented above, the analysis includes duration of stay in Spain as well as its squared term (henceforth also referred to as YIS – years in Spain). Whereas the descriptive statistics focus on the first ten years in Spain, the multivariate analysis takes into account the remittance behaviour irrespective of the duration of stay in Spain. In the absence of information on month of arrival, duration of stay is expressed as the number of calendar years the respondent has lived in Spain¹. To test the family-links hypothesis, we include a number of time-varying indicators of strength of ties to Senegal and Spain during each year in the destination country. More precisely, when it comes to ties to origin, we control for: 1) number of children in Senegal; 2) number of siblings in Senegal; 3) a dummy variable for respondents having a partner in Senegal; 4) a dummy variable that takes value 1 if the respondent's mother is alive and living in Senegal; and 5) a dummy variable that takes value 1 if the respondent's father is alive and living in Senegal. We also add three indicators of the strength of ties to Spain: 1) number of children in Spain, 2) number of siblings in Spain, and 3) having a partner in Spain. To test the autonomous migrant hypothesis, we include a dummy time-invariant variable that takes the value 1 if the respondent autonomously decides to migrate. As a means to test the pure material links hypothesis, we include dummy variables for 1) owning a house in Senegal, 2) owning a plot in Senegal, and 3) owning a business in Senegal. Including of a categorical variable for the labour market attachment allows us to test the capacity hypothesis. We distinguish between three categories: 1) non-employed (reference group), 2) employed, and 3) self-employed respondents. The physical links hypothesis is tested by including a dummy variable that takes the value 1 if the immigrant has visited Senegal at least once since arriving in Spain. Finally, age and its squared term, as well as education level, are also included in the model as control variables. Education is a time-invariant variable and refers to educational attainment at the time of the survey. The multivariate analysis is performed in two steps. Model 1 includes demographic variables, migration-specific variables (duration of stay in Spain and the nature of the migration decision-making process), and variables reflecting the strength of ties to origin. Model 2 complements Model 1 by also including variables for the immigrants' situation in the country in the destination country, namely labour market attachment and family ties to the destination country. As is the case in the descriptive analysis, the multivariate models are estimated separately for men and women.

Sample characteristics

Table 1 shows the characteristics of the sample at the time of the survey. Beginning with basic demographic characteristics, the average age of Senegalese migrants at arrival in Spain is about 28, for both men and women. Also for both sexes, more than half of the survey respondents have primary school or less. Turning to migration-specific variables, Senegalese men and women in Spain were fairly recent immigrants at the time of the survey, with an average duration of stay in the country at around 8.5 years. More than three quarters of men autonomously decided to migrate, whereas this is the case for only slightly more than one quarter of women. As for family ties to Senegal, gender differences are modestly pronounced along most dimensions. Men have on average 1.12 children left behind in Senegal, as compared to 0.75 among women. Regardless of gender, the Senegalese in our sample have an average of almost seven siblings living in Senegal. More than a half of the survey respondents have fathers living in Senegal. At the same time, around three out of four men and four out of five women have a mother living in Senegal. However, differences between men and women are clearly pronounced when it comes to partners. In particular, more than half of the men have a partner living in Senegal, whereas this is only the case for one out of eight women. The share of survey respondents who have a property in Senegal ranges between 18 and 29 per cent, depending on the type of property (plot or house) and gender (the share being somewhat higher among men). At the same time, a small percentage of respondents have a business running in Senegal at the time of the survey. More than two-thirds of men and

TABLE 1 Sample characteristics at the time of the survey (weighted)

	Men	Women
Age at arrival	28.19	27.58
Primary education or less	59.84	52.64
Secondary education	18.62	20.52
Higher education	21.54	26.84
Years in Spain	8.43	8.65
Migration own decision	77.67	26.18
No. of children in Senegal	1.12	0.75
Has partner in Senegal	56.48	12.37
No. of siblings in Senegal	6.62	6.87
Father alive	53.81	57.04
Mother alive	74.94	82.91
House in Senegal	28.79	20.23
Plot in Senegal	25.48	17.92
Business in Senegal	5.47	4.44
Has visited Senegal	50.57	46.83
No. of children in Spain	0.33	1.58
Has partner in Spain	14.42	66.84
No. of siblings in Spain	0.45	0.57
Not employed	30.58	50.92
Employee	47.19	34.34
Self-employed	22.22	14.74
N	302	305

Source: MAFE and MESE surveys

around half of women were employed at the time of the survey. Among both sexes, the share of employees was more than twice as high as that of the self-employed.

HOW MUCH AND WHEN DO THE SENEGALESE IN Spain REMIT?

Our initial analysis shows that most Senegalese migrants in Spain send money home at some point. More precisely, 73 per cent of men and 56 per cent of women remit at the time of the survey. In order to incorporate the time dimension into the analysis, we run Kaplan–Meier models of the initiation and termination of remittances. The first part of our analysis of the dynamics of remittance sending is presented in Figure 1, which shows how long after arrival it takes migrants to start remitting. Three main conclusions emerge from this analysis. First, a large majority of Senegalese in Spain send money home at some point. By the tenth year in Spain, around 90 per cent of men and around 70 per cent of women will have remitted at least once. Second, they start remitting fairly soon after arrival. Just above half of men and just above a third of women already start remitting in the first year after arrival. Finally, for each duration since arrival in Spain, the share of Senegalese migrants who have started remitting is somewhat higher among men. Moreover, a Cox regression-based test for equality of survivor functions suggests that differences between men and women are statistically significant. A separate analysis of hazard rates (detailed results not reported) shows that the likelihood to start remitting, conditional on not having started yet, decreases with the time spent in Spain. For instance, the likelihood for men to remit in the first year is 53 per cent, conditional on not having remitted yet. This likelihood falls to 14 per cent after five years and is only 3 per cent after ten years.

Figure 2 shows how much time it takes immigrants to stop remitting once they start. The results suggest, first, that the process of terminating remittances takes longer than it does to initiate a remittance flow and, second, that migrants are more likely to start remitting than to cease doing it. Only 7 per cent of men and 8 per cent of women stop remitting a year after the initiation of remittances. Even after ten years, three quarters of Senegalese immigrants will still be sending money back to Senegal. Also, gender differences are much less pronounced than they are for initiating remittances, and the test for equality of survivor functions finds no statistically significant difference between men and women when it comes to terminating remittances. Similar to what was found for the initiation of remittances, the likelihood to stop remitting decreases over time.

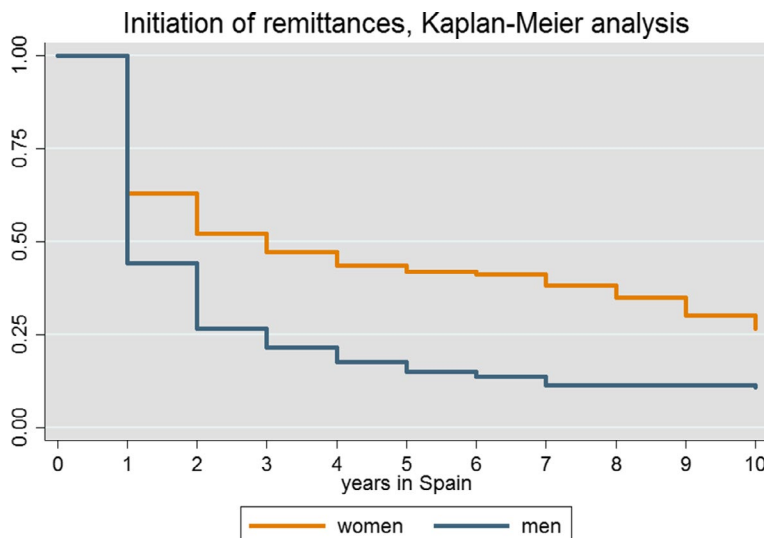


FIGURE 1 Source: MAFE and MESE surveys, own calculations

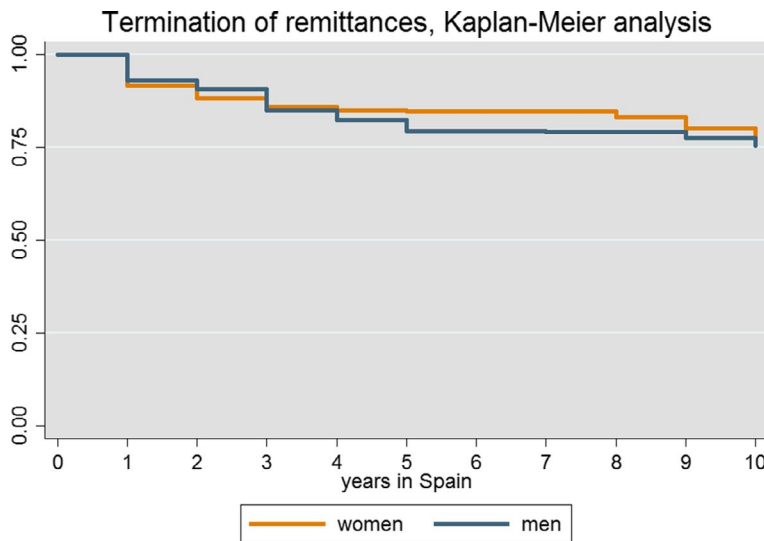


FIGURE 2 Source: MAFE and MESE surveys, own calculations

The two figures presented above and the results of the hazard rate analysis suggest that, for the same duration of stay, the likelihood to start remitting for the first time is generally higher than the likelihood to stop remitting. It thus comes as no surprise that, without controlling for observables, the association between remittance sending and duration of stay is positive during the first ten years in Spain, as can be seen in Figure A1 in the Appendix. Around 83 per cent of men and 68 per cent of women send money to Senegal during the tenth year of stay in Spain. In other words, although remittance sending is already substantial in the first year of stay in Spain, by the tenth year it becomes clearly dominant among women and nearly universal among men.

MULTIVARIATE ANALYSIS – WHO REMITS?

The results of the multi-level mixed effects models are shown in Table 2, separately for men and women. Beginning with the demographic predictors of remittance sending, the more parsimonious Model 1 shows that age at arrival is not a significant predictor of remittance behaviour for men. For women, the likelihood of remitting increases with age, although the relationship is non-linear. Some gender patterns also emerge when looking at education level, as the likelihood of remitting is 9 percentage points higher among men with tertiary education compared with men who have primary education. In contrast, no significant differences by education are found among women. Both duration of stay and its squared term are significant predictors of remittance behaviour. The signs and magnitudes of the coefficients suggest that, after controlling for observables, the association between duration of stay and remittances is non-linear. More precisely, the likelihood of remitting initially increases with time spent in Spain, and it reaches its maximum after 8 years for men and 11 years for women. The family links hypothesis receives some support from our model. Having children increases the chances of remitting, and more so among women. Among men, having a partner living in Senegal increases the chances of remitting, whereas having siblings in Senegal makes remitting more likely among women. The analysis shows little support for the autonomous migrant hypothesis, as the association between the type of migration decision-making and remittances is not statistically significant among men. However, this is not the case for women, among whom the likelihood of remitting is almost 13 percentage points higher if they autonomously decided to migrate as compared to those whose migration was the result of a collective decision. Finally, Model 1 supports neither the material links hypothesis nor the physical links hypothesis, as property in and visits to Senegal have no significant associations with remittances.

TABLE 2 Remittance sending among Senegalese migrants in Spain, mixed-effects linear probability regression

	Men		Women	
	Model (1)	Model (2)	Model (1)	Model (2)
Age at arrival	0.017	0.018	0.031**	0.020
Age at arrival sq.	-0.000	-0.000	-0.001**	-0.000
No education or primary (ref.)				
Secondary	-0.140	0.002	0.054	0.046
Higher education	0.089*	0.067	0.024	0.003
YIS	0.046***	0.038***	0.051***	0.044***
YIS ² /100	-0.290***	-0.196***	-0.241***	-0.198***
Migration own decision	0.005	0.007	0.126***	0.104**
No. of children in Senegal	0.021***	0.021*	0.030***	0.035***
Has partner in Senegal	0.047***	0.037*	0.014	0.005
No. of siblings in Senegal	-0.000	0.002	0.011**	0.008
Father alive	0.046	0.025	0.004	-0.012
Mother alive	0.006	-0.011	0.014	0.001
House in Senegal	0.046	0.034	0.053	0.061*
Plot in Senegal	0.029	0.034	0.044	0.055
Business in Senegal	-0.070	-0.051	0.012	-0.016
Has visited Senegal	-0.001	-0.003	0.024	0.019
No. of children in Spain		0.002		0.001
Has partner in Spain		-0.013		0.018
No. of siblings in Spain		0.034**		-0.040***
Not employed (ref.)				
Employed		0.294***		0.204***
Self-employed		0.249***		0.191***
Constant	0.187	-0.049	-0.274	-0.133
Between-individual variances/covariances				
Intercept variance	0.243 s.e. 0.017	0.205 s.e. 0.016	0.224 s.e. 0.014	0.190 s.e. 0.013
YIS variance	0.014 s.e. 0.003	0.010 s.e. 0.002	0.010 s.e. 0.002	0.009 s.e. 0.002
YIS ² /100 variance	0.480 s.e. 0.209	0.306 s.e. 0.145	0.357 s.e. 0.140	0.272 s.e. 0.085
Intercept-YIS covariance	-0.039 s.e. 0.006	-0.030 s.e. 0.005	-0.020 s.e. 0.004	-0.018 s.e. 0.004
Intercept-YIS ² /100 covariance	0.179 s.e. 0.050	0.126 s.e. 0.038	0.057 s.e. 0.031	0.059 s.e. 0.025
Covariance (YIS, YIS ² /100)	-0.076 s.e. 0.025	-0.053 s.e. 0.019	-0.055 s.e. 0.017	-0.046 s.e. 0.011
Within-individual variance	0.034 s.e. 0.003	0.033 s.e. 0.003	0.034 s.e. 0.003	0.033 s.e. 0.003
N	302		305	
Person-years	3,091		3,041	

Source: MAFE and MESE surveys; * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$

Model 2 introduces variables indicating the capacity to remit and family ties to Spain. As expected by the capacity hypothesis, attachment to the labour market increases the likelihood of sending remittances. Moreover, the effect of employment is sizeable and surpasses that of other significant variables. Compared to the non-employed and all else being equal among Senegalese men, the likelihood of sending remittances is 25 percentage points higher among the self-employed and 29 percentage points higher among employees. The corresponding figures for women who are self-employed and employees are, respectively, 19 and 20 percentage points. In contrast to the role of economic ties to Spain, the importance of family ties to the country of destination is fairly weak. Having children or a partner in Spain is not associated with remittance sending, neither for men nor for women. For men, each sibling residing in Spain adds 3.4 percentage points to the likelihood of remitting. Interestingly, we find the opposite for women, as each sibling living in Spain lowers the likelihood of sending remittances by 4 percentage points.

Including new variables in Model 2 brings about small to moderate changes in the magnitudes of most variables from Model 1. The most substantial change among men is the disappearance of significant associations between being highly educated and sending money to Senegal. Among women, age and the number of siblings in Senegal are no longer significant predictors of remittances. However, homeownership becomes significant at the 10 per cent level, with the coefficient suggesting that owning a house in Senegal adds 6 percentage points to the likelihood of remitting. We obtain a similar coefficient for having a plot in Senegal; however, this result only comes close to but does not reach the 0.1 significance level ($p < 13$ per cent). The lower part of the table shows random effects. Once we adjust for all the covariates (Model 2), between-individual variance is estimated at 0.205 for men and 0.190 for women, whereas within-individual variance for both sexes is 0.033. This implies that, among men, 86 per cent of the total residual variance in remitting behaviour is due to differences between individuals, while the remaining 14 per cent are ascribed to within-individual variance over time. The corresponding shares among women are 85 per cent and 15 per cent.

In order to further address the question of differences in remittance behaviour between men and women, we implemented an additional regression by including both sexes and a dummy variable for men (details not reported). The results show that, once observables are taken into account, the likelihood of remitting is 8 percentage points higher among men.

CONCLUSION

We use data from two retrospective surveys on Senegalese migrants in Spain to explore the patterns of remittance sending from destination to origin. Motivated by strong gendered patterns in migrations between Senegal and Europe, we perform a separate analysis for men and women. Using a longitudinal approach, we find that around 90 per cent of men and 70 per cent of women will have remitted at least once by the end of their tenth year in Spain. Remittance sending becomes substantial already in the first year of arrival. In addition, three quarters of the great majority of Senegalese who end up sending remittances maintain this economic flow for at least ten years, suggesting the presence of sustained transnational ties over time. The high prevalence of remittance sending in this paper is in line with the remittance behaviour of Senegalese in France and Italy, as documented by Chort et al. (2012). It also matches the aggregate-level statistics on remittance sending among Senegalese living abroad. However, it should also be noted that our analysis indicates a non-linear inverse U-shaped association between time in the destination country and remittances, once observable characteristics are taken into account. More precisely, the probability of remitting peaks in the eighth year in Spain for men and in the eleventh year for women. This non-linear association is in accordance with most previous studies from other migration contexts.

Drawing on a strong link between labour market participation and remittances for both sexes in our multivariate analysis, we argue that strong support exists for the capacity hypothesis. The analysis lends some support to the family-links hypothesis as well, mainly through a positive association between having children in Senegal

and remittance sending. This association is found among both men and women. Having a partner in Senegal also increases the likelihood of remitting, but only among men. This finding may reflect Senegal's patriarchal norms that are dominant in both sexes, according to which economic responsibility falls primarily on men (Adjamagbo et al., 2006; Pilon & Vignikin, 1996). Concerning the material-links hypothesis, we find no evidence for it among men, whereas among women it is supported through a positive significant association between owning a house in Senegal and remittance sending. In the Senegalese context, international migration helps overcome gender inequalities in access to asset ownership (Mezger & Beauchemin, 2015). It is thus possible that asset ownership has a stronger symbolic meaning for Senegalese women than for men, which may explain the gender difference related to the material links hypothesis. We find no support for the physical-links hypothesis, as the analysis shows no link between visits to Senegal and remittances. The autonomous migrant hypothesis is also the only gendered hypothesis, as it concerns only men. However, as we find no link between migration being the migrant's own decision and subsequent remittances, this hypothesis is not supported either. Remittance sending among Senegalese men in Spain is almost universal, and it may be the case that remittance-related norms and expectations in Senegal are such that the commitment to remit is not weaker among autonomous movers. On the other hand, our analysis shows that, all else being equal, women who migrated on their own initiative remit more. Considering that remittance dyads (sender-to-receiver pairings) are not gender neutral in immigrant families from male-dominated societies (given that the man and his family take priority, King et al., 2013), this finding may suggest that female agency in the migration decision-making process is positively associated with women's bargaining power when deciding who in the family remits to whom in Senegal.

Family ties to destination are weak predictors of remittance behaviour in our analysis, with only the number of siblings in Spain having been found to be significantly correlated with our dependent variable. The number of siblings in Spain increases remittances among men, which may suggest that a stronger family presence exerts some pressure on men to send remittances. In contrast, a negative association between the two variables among women may suggest that a strong sibling presence increases the likelihood that remittances are expected from other family members, namely brothers, thus freeing female immigrants from this obligation. This explanation would be in line with Liu et al. (2018), who find that sibling networks influence Senegalese intra-African migration while it is specifically brother networks that motivate emigration to Europe. Interestingly, we find no link between remittances and having children in Spain. This result may be indicative of Senegalese immigrants' strong ties to their extended families in the home country.

The separate analyses for men and women in our paper were motivated by strong gender norms in Senegal, particularly by pronounced gender patterns in migration behaviour. Based on our results, it can be argued that very different migration mechanisms for men and women produce moderate differences in remittance behaviour. Men are somewhat more likely to remit overall. The event history analysis reveals that they are more likely than women to start remitting. On the other hand, once they start remitting, women are as likely as men to maintain their remittances. Men's propensity to remit remains somewhat higher even after controlling for observables. This difference may be due to the expectations of the patriarchal home community, but it is also possible that economically active Senegalese men in Spain have a higher income and thus more resources than economically active women. Multivariate analyses by gender reveal that most coefficients for men and women in the full model are similar in terms of significance level, sign, and magnitude. The random effects obtained in separate estimates by gender are very similar as well. Once we control for observables, as much as 86 per cent of the residual variance for men and 85 per cent for women are due to differences between individuals, with the remaining share being due to within-individual variance over time. However, although few gender differences were found, those that we have identified and discussed above are important for understanding remittance behaviour and transnational ties, and they certainly deserve to be scrutinized in future research.

This study is not without limitations. As is often the case in the literature on remittances (see Carling, 2008), we know only whether or not a person remits while the actual amounts of money sent remain unknown. Using the MAFE household survey, Diagne and Rakotonarivo (2010) studied the amounts and frequency of

remittances obtained by the households from the Dakar region. Their results show that 57 per cent and 10 per cent of migrants remitted, respectively, 100,000 and 200,000 West African CFA francs (FCFA) or less per transfer (roughly equivalent to 150 and 300 euros at the time of their study). For the sake of comparison, the average monthly salary in the Senegalese formal private sector was 220,000 FCFA in 2009, and the average monthly public sector salary was 133,000 FCFA (ILO, 2013). They also found that, among those who remit, four out of ten do so at least once a month. Another limitation of our study is that we cannot observe either the recipients or purposes of the money being sent. Finally, although our analysis suggests that financial capacity plays a very important role in remittance sending, we are only able to control for level of attachment to the labour market and have no reliable information on earnings or other sources of income. A natural progression of this work in years to come would be to run a similar survey of Senegalese immigrants in Spain and analyse potential changes in patterns of remittance sending. At the time of the two surveys used in our analyses, Spain was still emerging as a destination country and, consequently, most of the Senegalese in our sample were recent immigrants to Spain. They have since become a more established community while the destination country has itself undergone substantial socio-economic changes.

ACKNOWLEDGEMENTS

The paper benefited from the comments received at the European Population Conference in Brussels in 2018. The authors are also thankful for the comments made by two anonymous reviewers.

PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1111/imig.12997>.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in <https://mafeproject.site.ined.fr/en/>.

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ENDNOTE

1. The lowest value of this variable is 1. For instance, if a respondent arrived in Spain in 2004, the variable duration of stay takes value 1 in 2004, value 2 in 2005, etc.

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How to cite this article: Carrasco, J.I. & Obućina, O. (2022) The dynamics of remittance behaviour among Senegalese men and women in Spain. *International Migration*, 00, 1–17. Available from: <https://doi.org/10.1111/imig.12997>

APPENDIX 1

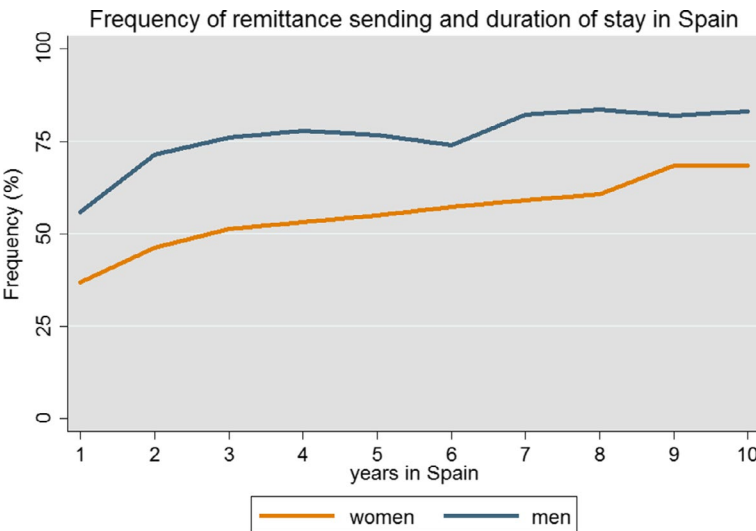


FIGURE A1 Source: MAFE and MESE surveys, own calculations